

WEEK 4 ASSIGNMENT

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

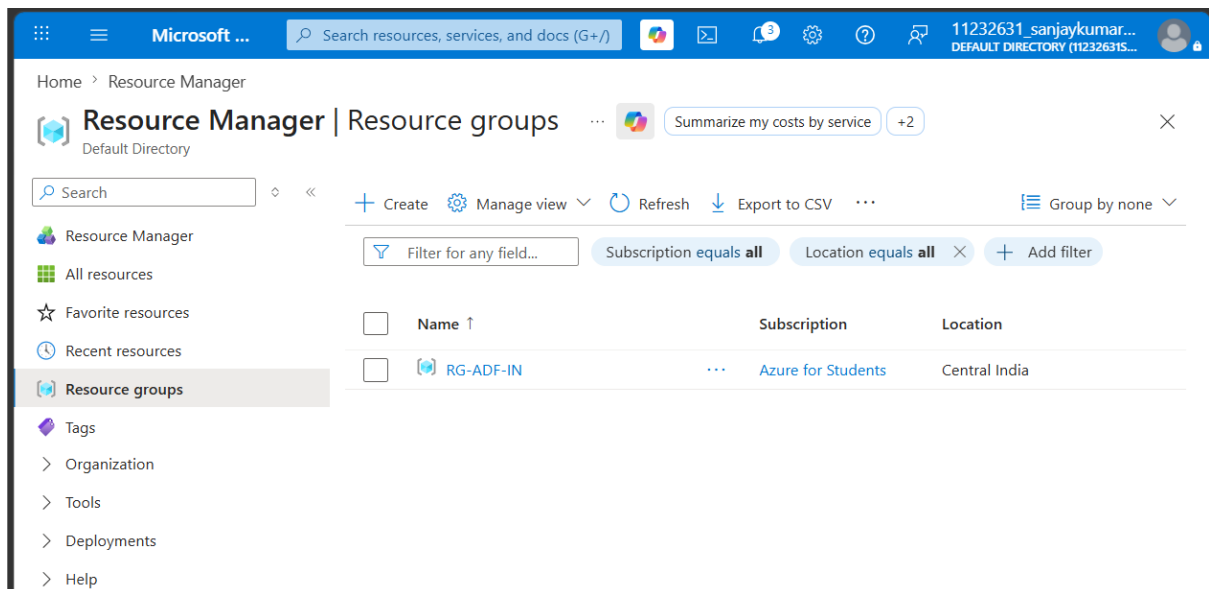
To understand Azure cloud concepts and build a complete data pipeline using Storage Account

and Azure Data Factory.

Assignment Tasks

Task 1:

- Explored Azure Portal
- Created a Resource Group



Task 2: Storage Setup

- Create a Storage Account

Microsoft ... Search resources, services, and docs (G+/) 11232631_sanjaykumar... DEFAULT DIRECTORY (11232631S...

Home > Resource Manager | Resource groups > RG-ADF-IN

adfstorageeci

Storage account

Show me alerts for this storage account Create an LCM rule for my storage account +1

Search Upload Open in Explorer Delete Move Refresh Open in mobile

Overview

- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events
- Storage browser
- Storage Mover
- Partner solutions
- Resource visualizer
- Data storage
- Security + networking
- Data management
- Settings
- Monitoring
- Automation
- Help

Essentials

Resource group (move) [RG-ADF-IN](#)

Location: centralindia

Subscription (move) [Azure for Students](#)

Subscription ID: e10f9dbd-f5d4-45c5-84b9-3cc5074946f9

Disk state: Available

Tags (edit) [Add tags](#)

Performance: Standard

Replication: Locally redundant storage (LRS)

Account kind: StorageV2 (general purpose v2)

Provisioning state: Succeeded

Created: 6/14/2026, 12:54:03 PM

Properties

Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

Hierarchical namespace: Disabled

Default access tier: Hot

Blob anonymous access: Disabled

Blob soft delete

Security

Require secure transfer for REST API operations: Enabled

Storage account key access: Enabled

Minimum TLS version: Version 1.2

Add or remove favorites by pressing Ctrl+Shift+F

- Create a Blob Container

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adfstorageeci | Containers

Storage account

Search Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix Only show active containers

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	\$logs	6/14/2026, 12:54:28 PM	Private	Available	-
☐	destination-data	6/14/2026, 1:13:38 PM	Private	Available	-
☐	source-data	6/14/2026, 12:56:40 PM	Private	Available	-

- Upload a CSV file

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source-data

Container

Search Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive) Only show active blobs

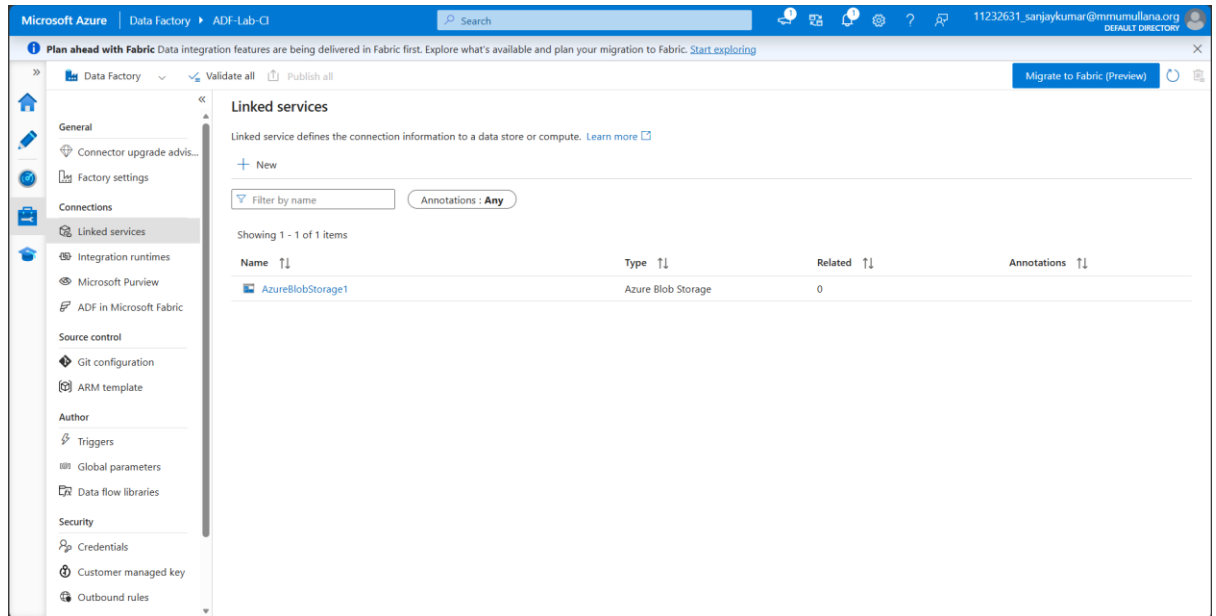
Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Sample - Superstore.csv	6/14/2026, 1:00:10 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

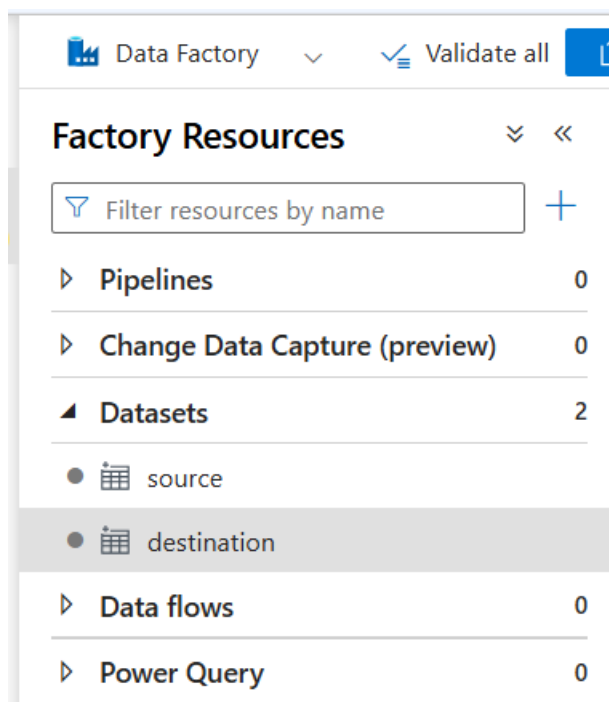
Task 3: ADF Basics

Create Azure Data Factory

- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)



- Create datasets (source + destination)



- Use Get Metadata activity

The screenshot shows the Microsoft Azure Data Factory console. The pipeline 'SuperstorePipeline' is displayed with two activities: 'Get Metadata' (Get Metadata1) and 'Copy data' (Copy data1). The pipeline status is 'Succeeded'. The output table shows the following details:

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime	User property	Activity run ID
Copy data1	Succeeded	Copy data	6/14/2026, 1:36:01 PM	15s	AutoResolveIntegrationRuntime (Central India)	3c115eef-4aef	
Get Metadata1	Succeeded	Get Metadata	6/14/2026, 1:35:46 PM	14s	AutoResolveIntegrationRuntime (Central India)	2d0d610d-84e	

Task 6: IAM Roles

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

The screenshot shows the Access Control (IAM) settings for the storage account 'adfstorageeci'. The roles assigned are:

- Owner (2):**
 - Sanjay Kumar Sah (d3f8afd0-3009-4c07-83be-8...) - User - Owner - Subscription (Inherited)
 - Sanjay Kumar Sah (d3f8afd0-3009-4c07-83be-8...) - User - Owner - Subscription (Inherited)
- Contributor (1):**
 - Sanjay Kumar Sah (d3f8afd0-3009-4c07-83be-8...) - User - Contributor - This resource
- Reader (1):**
 - Sanjay Kumar Sah (d3f8afd0-3009-4c07-83be-8...) - User - Reader - This resource
- Storage Blob Data Contributor (1):**
 - ADF-Lab-CI (be26d4f3-3596-4d82-a47f-...) - Managed identity - Storage Blob Data Contributor - This resource

Showing 1 - 5 of 5 results.

Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully

The screenshot displays the Microsoft Azure Data Factory interface. The left sidebar shows the 'Activities' pane with a search bar and a list of activity types: Move and transform, Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, General, HDInsight, Iteration & conditionals, Machine Learning, and Power Query. The main workspace shows a pipeline named 'SuperstorePipeline' with two activities: 'Get Metadata1' and 'Copy data1'. The 'Copy data1' activity is highlighted, showing its configuration. Below the activities, the 'Output' tab is selected, displaying the pipeline run details. The pipeline run ID is '8f8bd9b4-6cdc-4189-9cec-3fe4a6a8d7b0'. The pipeline status is 'Succeeded'. The table below shows the execution details for the activities.

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime	User property	Activity run ID
Copy data1	Succeeded	Copy data	6/14/2026, 1:36:01 PM	15s	AutoResolveIntegrationRuntime (Central India)		3c115eef-4aef
Get Metadata1	Succeeded	Get Metadata	6/14/2026, 1:35:46 PM	14s	AutoResolveIntegrationRuntime (Central India)		2d0d610d-84f

- Data copied to destination

The screenshot displays the Microsoft Azure portal interface. The left sidebar shows the 'destination-data' container. The main workspace shows the 'destination-data' container details. The 'Overview' tab is selected, showing the container's authentication method as 'Access key'. The table below shows the blobs in the container.

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	6/14/2026, 1:36:13 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

- Metadata validated



Output



Copy to clipboard

```
{
  "exists": true,
  "size": 2287806,
  "lastModified": "2026-06-14T07:30:10Z",
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 8
  },
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": [
      {
        "meterType": "AzureIR",
        "duration": 0.016666666666666666,
        "unit": "Hours"
      }
    ]
  }
}
```